

# Harvard College Debating Union Season Opener 2026- Aff Case Doc

## Aff

The US is a declining empire, but AI can be our saving grace. It can ignite growth and restore U.S. dominance, yet unchecked expansion threatens our grid, our economy, and the communities that host these centers.

A data-center moratorium strikes the necessary balance: stopping reckless development without surrendering American progress.

History is on our side. After initial growing pains, the Netherlands used a moratorium to give its data center hub renewed attraction and a clear path to sustainable growth. In Singapore, a temporary pause pushed providers toward renewables and smarter strategies, and industry experts concluded it would not deter long-term investment. And these pauses are temporary: Amsterdam lifted its ban after just one year, once clear zoning and power limits were in place.

To ensure AI secures America's future without destroying its communities, UNI Institute affirms.

### **Our sole contention concerns Strengthening Our Foundations.**

Despite AI hype being at an all-time high, the future isn't promising. **Senior Analyst Tedder in 2026 writes that** even if the US leads AI, that lead will be difficult to sustain if our physical infrastructure can't keep pace. China's centralized system lets it build ahead of demand, while the US is slowed by fragmented regulation and transmission bottlenecks. By 2030, China's power generation additions will be 4x those of the US. China isn't just catching up; it is building momentum in real-world adoption.

A pause on data centers strengthens our foundations in two specific ways:

#### ***The first is clearer regulations.***

In the US, each state decides how it wants to regulate AI, leading to a patchwork of confusing laws. **Fortune author Richardson in 2026** finds that states introduced over 1,200 AI-related bills last year, creating contradictory rules that multiply compliance burdens. A hiring tool that satisfies California must separately satisfy Colorado and New York City. This patchwork adds 17% to AI system costs, and Harvard researchers find rising compliance costs can flip a startup's operating margin from positive 13% to negative 7%: the difference between survival and bankruptcy. Meanwhile, Chinese AI companies operate under one unified national framework.

A moratorium forces federal action. **Sokler of Mintz in 2026** explains that the AI Data Center Moratorium Act would impose a nationwide pause on the construction and expansion of AI data centers until Congress enacts comprehensive federal safeguards.

Growth resumes the moment Congress acts, so the pause turns today's gridlock into a deadline.

And that pause buys policymakers time. **Etheridge of the International Center for Law & Economics** writes that freezing the patchwork preserves policymakers' ability to take stock and design a unified national framework, while the current wave of state bills risks locking in conflict.

Otherwise, the patchwork is devastating. **Castro of ITIF** finds that a patchwork of state tech laws would cost over \$1 trillion over ten years, with \$200 billion hitting small businesses.

### ***Second is strengthening our grids.***

AI will overload the electricity grid. **Energy reporter Behr in 2026** finds that the grid watchdog NERC calls AI's extreme power swings, hundreds of megawatts in an instant, a "high likelihood, high impact" risk to a grid already under stress from extreme weather. Center construction outpaces grid upgrades, and centers don't tell operators when these shifts are coming. NERC warns that a cluster of supersize data centers dropping off the grid would create an unacceptable risk of cascading shutdowns and widespread outages. 50 scientists from NVIDIA, Microsoft, and OpenAI document the risk.

Affirming allows time for the grid to recover. **Author Kelly from the NRECA** finds that centers are developed faster than the infrastructure needed to support them, threatening reliability. The significant challenge is the size and speed at which centers are built: often within two years, much faster than traditional industrial loads. A pause attacks that speed directly.

And, a moratorium forces data center companies to invest in the grid. **New York Governor Kathy Hochul** writes that New York's moratorium holds new data centers to consistent standards and moves to require them to invest in grid infrastructure, and it lifts once those standards are finalized.

Without a strong grid, AI development is impossible. **The International Energy Agency** finds that there is no AI without energy, and data center electricity demand will more than double by 2030. The energy sector is the heart of one of the most important technologies today.

Overall, letting the foundations for long-term AI growth weaken would cost us the economy of the future. **Sheehan from the University of San Diego** writes that AI has the potential to grow from a \$2 trillion to a \$15.7 trillion market by 2030. Strong foundations decide whether America captures that growth or hands it to China.

When our full-steam-ahead strategy for growth sets us up for future failures, you know something has to change. You know you have to affirm.

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## **Framing: Moratoriums Work**

## Netherlands proves: after early growing pains, Amsterdam's moratorium created renewed attraction and a clear path for sustainable industry growth

**Derksen '21**, Door Zoe Derksen, 6-21-2021, Two years after the Amsterdam moratorium, where does its data center industry stand?, Dutch Data Center Association, <https://www.dutchdatacenters.nl/en/nieuws/two-years-after-the-amsterdam-moratorium-where-does-its-data-center-industry-stand/> //RonitS

Back **in 2019**, the **moratorium** has **caused** serious **damage to the sector**. Being the first to announce such a moratorium, Amsterdam sent a message to investors: data centers are not welcome here.

**However, two years later, Amsterdam has gained a whole new role in the global data center realm, with expertise on** how it is considering the industry in terms of **sustainability**.

**With the new policies implemented, the data center hub has gained renewed attraction.** We have seen recently that more governments, such as Singapore, Frankfurt, some regions in China and Ireland, have been forced to go down this road. By the time the issues elsewhere are solved, the Netherlands is likely to have more dedicated areas for data center development and policies that provide a clear path for industry growth.

More sustainable data center builds are an undeniable part of our future. One example of this is the European Green Deal in the form of the Climate Neutral Data Centre Pact, which requires data centers to be climate neutral in 2030. With the new sustainable policies, it makes sense that **the Netherlands' still beholds a promising future**. We therefore **don't expect any Friday summer days' surprises in the coming years.**

## Singapore too: the pause forced providers toward renewables and better strategies without deterring long-term investment, and Amsterdam lifted its ban after one year

**Kit '21**, Tang See Kit, 5-10-2021, "Singapore puts 'temporary pause' on new data centres: Why and what it means for the industry," CNA, <https://www.channelnewsasia.com/business/new-data-centres-singapore-temporary-pause-climate-change-1355246> //RonitS

"QUALITY, NOT QUANTITY"

There has been a "rapid increase" in the number of **data centres** being approved to be constructed on industrial state land over the past five years, MTI said in a separate written parliamentary reply on Feb 1.

Fourteen data centres with a total IT capacity of 768 megawatts **were** given the green light during that period, compared to 12 data centres carrying 307 megawatts in the preceding five years.

With the industry being **an "intensive" user of electricity and water**, a **"temporary** pause" **has** since **been imposed** on the building of new data centres **as local authorities mull a balance between environment sustainability and supporting business needs.**

"The Government therefore decided to moderate the growth of data centres with a temporary pause on the release of state land for data centres, as well as the development of data centres on existing state land," MTI's reply said.

It added that the industry was informed of the decision in 2019 and has been consulted on ideas for a more sustainable growth.

"We will **strive for quality, not quantity** – this means we **seek to anchor a range of data centres that can meet both industry and society's needs, are best in class in terms of resource efficiency, and that continuously innovate to push the boundaries of resource efficiency** ... in a tropical climate," MTI said.

The Government has said it intends to share its plans later this year after concluding its review.

Industry experts told CNA that the Government's decision comes as no surprise, given the country's land and power constraints.

Singapore is not alone in doing so. Amsterdam, the capital of the Netherlands, hit the brakes on new data centres in July 2019, citing strains on its property market and power networks. **The ban was lifted after a year under conditions such as locating data centres in specific zones with limits on power use** and floor space.

"Singapore is a relatively smaller city-country, when compared to the other tier-1 markets such as Tokyo, Sydney and Hong Kong. Yet we come in second in terms of IT capacity," said Ms Lim ChinYee, senior director of Asia-Pacific data centre solutions at CBRE.

"It is therefore **important that Singapore ensures that the overall data centre ecosystem is environmentally sustainable, while balancing the need for growth.**"

Mr Farrell thinks this **offers Singapore "a timely opportunity" to invest in green technologies, higher-performance infrastructure and develop technology talent.**

"This will **help the country keep up with future expansion in data centre demand** when the moratorium is lifted," he said.

ANY IMPACT?

But with supply being curbed at a time of booming demand, rental rates at data centres based on cost per kilowatt have surged by more than 30 per cent last year, said Mr Lynus Pook, Colliers' senior director of industrial advisory in Asia.

These rates have stayed low from 2016 to 2019, especially in 2018 when vacancy rates peaked at 40 per cent due to new supply.

"With the temporary pause in building new data centres, the vacancy rates have narrowed to 5 per cent in 2020, leading to some spill-over in demand to neighbouring countries such as (Malaysia) Johor Bahru," said Mr Pook. "Inevitably, rental rates have increased by more than 30 per cent."

Ms Lim expects rental rates to continue heading north over the next three years, given a tight supply and expectations that the moratorium on constructing new data centres will be lifted by 2022.

**Singapore has been a popular site to build data centres** due to factors such as a reliable power grid and a business-friendly environment. **So far, it remains the case even with the ongoing moratorium, industry observers said.**

For instance, **Singapore came in second in a recent list of top data centre markets** by Arcadis, coming **just behind the United States**, due to political stability, a well-educated workforce and the ability to attract major technology investments.

**Rather than the moratorium being considered off-putting for potential data center providers, it has forced them to rethink their current strategies and look for alternative models, as they have already started to do globally anyway,"** said Mr Richard Warburton, growth director for Southeast Asia at Arcadis.

He pointed to announcements by companies such as Facebook and Amazon that will be tapping renewable energy to power their data centres in Singapore.

"Overall, **it's unlikely that the moratorium will deter providers from investing in Singapore in the mid to long term.**" said Mr Warburton, adding that it is "only natural" for data centre operators to move into regional markets as part of their expansion plans.

"Alternative data centre hubs are emerging across the region in response to increasing demand and corporate strategies, rather than as a reaction to the moratorium," he added.

## Contention 1: Strengthening Our Foundations

The US is losing its AI lead to China: fragmented regulation and lagging power infrastructure are the bottleneck

**Tedder '26**, Ben Tedder, 6-29-2026, "Listed Infrastructure: The US-China AI race may hinge on US power markets," Morrison Global, <https://morrisonglobal.com/news-insights/listed-infrastructure-the-us-china-ai-race-may-hinge-on-us-power-markets/> //RonitS

AI is often discussed as a contest over semiconductors, models and software. Increasingly, however, it is also a contest over power. The compute required to train and run frontier models is driving a surge in data centre demand, and that is turning power infrastructure into a strategic constraint. This matters because although the US still leads in many of the core technologies underpinning AI, that lead will prove difficult to sustain if the physical infrastructure required for deployment cannot keep pace.

CHINESE AI CAPABILITIES ARE NOT FAR BEHIND THE US, AND AI USAGE IS MUCH HIGHER IN CHINA China is using its vast domestic market to scale AI quickly across consumer and enterprise applications, helped by low-cost, open-source models and tightly integrated platforms. At the same time, improving domestic chip capability is helping narrow the compute gap with the West, increasing GPU self-sufficiency and lifting large language model performance. Importantly, competitiveness is increasingly being judged not just by peak chip performance, but by total cost per unit of compute and the ability to deploy at scale. That shift matters because it plays to China's strengths in low-cost deployment and rapid commercialisation. The sharp rise in token consumption among leading Chinese models in 2026 suggests usage is already scaling quickly, reinforcing the view

that China is not just catching up technologically, but also building momentum in real-world adoption.

### **CHINA'S PLANNED GENERATION BUILD-OUT DWARFS THE US**

This is where the US-China gap becomes most tangible, given the strategic imperative for each country to develop and control its own AI capabilities. While both countries face rising power needs from AI, China appears structurally less constrained in its ability to supply the required power infrastructure, aided by a more centralised system and a greater willingness to build ahead of demand. In the US, by contrast, growth is often slowed by fragmented regulation, lengthy interconnection queues, transmission bottlenecks and uncertain cost allocation. Indeed by sheer scale, Chinese power generation additions are expected to be 4x that of the US by 2030, with China's installed generated capacity projected to grow c.60% over the period, vs 25% growth in the US<sup>1</sup>. If AI is becoming a strategic industry, then resolving roadblocks to power and transmission build-out is no longer just an energy policy issue; it is part of the wider effort to ensure the US remains competitive in the AI race.

### **Subpoint A: Clearer Regulations**

In the US, each state decides how to regulate AI, creating a contradictory patchwork that adds 17% overhead, bankrupts startups, and hands China the advantage

**Richardson '26**, James Richardson and Eric Tanenblatt, 1-30-2026, "America's AI regulatory patchwork is crushing startups and helping China," Fortune, <https://archive.is/20260130132649/https://fortune.com/2026/01/30/americas-ai-regulatory-patchwork-is-crushing-startups-and-helping-china/> //RonitS

Last year, states introduced more than 1,200 AI-related bills, with at least 145 becoming law, creating contradictory requirements that multiply compliance burdens. Each jurisdiction defines "artificial intelligence," "high-risk systems," and "consequential decisions" differently, forcing companies to analyze identical technology under multiple incompatible frameworks. A hiring tool that satisfies California's four-year recordkeeping and anti-bias testing requirements must also meet Colorado's separate impact assessment mandates and New York City's independent bias audit regime with distinct notice requirements. Each jurisdiction defines core terms differently, forcing companies to analyze identical systems under multiple

incompatible frameworks. Industry estimates suggest **compliance costs add approximately 17 percent overhead to AI system expenses**. For small businesses, California's privacy and cybersecurity requirements alone impose nearly \$16,000 in annual compliance

costs. But these figures understate the true burden because they treat compliance as a variable cost that scales with company size. The reality is far worse. Harvard Kennedy School researchers identified what they termed a "compliance trap" in which

**regulatory costs consume resources faster than startups can generate revenue**.

Their analysis found that a 200 percent increase in fixed compliance costs transforms a startup's

operating margin from positive 13 percent to negative 7 percent. That's not a rounding error, **it's the**

**difference between survival and bankruptcy**.

A three-person team building an employment screening tool faces the same baseline compliance obligations as a thousand-person enterprise in many jurisdictions, but without the revenue base or legal infrastructure to absorb those costs. This dynamic hands an enormous competitive advantage to the very companies state regulations purport to constrain. Incumbent tech giants maintain compliance departments that dwarf entire startups. They can afford multi-jurisdictional legal teams, custom bias auditing frameworks, and the political relationships necessary to shape emerging requirements. For them, state AI fragmentation represents a manageable cost of doing business. For startups, however, it represents an insurmountable barrier to entry. The strategic implications are staggering. **While**

**American entrepreneurs waste engineering talent on contradictory compliance regimes, Chinese AI companies operate under a unified national framework**.

Beijing's approach is hardly a model of light-touch regulation, but it offers what America's

patchwork cannot: coherent rules that don't change at state lines. **When compliance costs**

**exceed development budgets, innovation doesn't slow—it stops altogether or**

**moves to foreign jurisdictions where the rules are clear**. The White House recognized

this competitive danger in a December 2025 executive order criticizing the "patchwork of 50

different regulatory regimes" and directing the U.S. Department of Justice to establish an AI Litigation Task Force to challenge state

laws that obstruct national AI policy. The order represents a necessary first step, but executive action alone cannot solve a problem rooted in legislative fragmentation. **Congress must act**. Federal preemption legislation would establish uniform national standards for AI systems while

preserving states' ability to enforce general consumer protection laws. This isn't a radical concept. Federal frameworks already govern aviation safety, pharmaceutical approvals, and telecommunications—industries where state-by-state variation would create identical chaos. Safe harbor provisions could protect companies that implement reasonable bias testing and impact assessments from liability, creating incentives for responsible development without imposing contradictory mandates. **The current trajectory is unsustainable. Every month that passes**

**with this regulatory chaos intact represents another month of American**

**innovation surrendered to better-organized competitors**. State legislators designed their AI frameworks

to constrain Big Tech's power. Instead, they've built a moat around incumbents while crushing the startups that might challenge them. The

compliance trap isn't protecting consumers. It's protecting monopolies and handing a

competitive advantage to foreign adversaries. That's not regulation, it's self-sabotage.

## **A moratorium forces federal action: the AI Data Center Moratorium Act pauses construction until Congress enacts comprehensive federal safeguards**

**Sokler '26**, Bruce D. Sokler, Alexander Hecht, Erik L. Barron, Christian Tamotsu Fjeld, Nicole Y. Teo, 3-31-2026, "White House Releases National AI Legislative Framework, as Competing Congressional Proposals Sharpen the Federal-State Divide — AI: The Washington Report," Mintz, <https://www.mintz.com/insights-center/viewpoints/54731/2026-03-31-white-house-releases-national-ai-legislative-framework> //RonitS

On March 20, the White House released the National AI Legislative Framework, offering a nonbinding legislative roadmap of President Trump's policy priorities in shaping a federal framework for AI legislation. The latest framework was issued pursuant to President Trump's December EO to establish a coordinated federal effort to block or override burdensome state AI regulations. By outlining policy goals for Congress, the framework seeks to operationalize the Trump administration's goal of creating a federally centralized and innovation-focused framework for AI governance. This framework attempts to address the policy tension that we have noted throughout the last year of this newsletter: should AI governance be primarily federal, or continue to evolve through state-led initiatives and legislation? Senator Marsha Blackburn (R-TN) has taken a leading role in advancing this agenda, releasing a discussion draft, the TRUMP AMERICA AI Act, on March 18, that would codify elements of the December EO and align closely with the

White House framework. The proposal incorporates provisions addressing online safety, intellectual property, and platform accountability, drawing on existing bipartisan measures such as child safety and digital likeness protections.

Five days after President Trump released the AI Framework, Sen. Bernie Sanders (I-VT) and Rep. Alexandria Ocasio-Cortez (D-NY-14) introduced **the AI Data Center Moratorium Act**, which **would impose a nationwide pause on the construction and expansion of AI data centers until Congress enacts comprehensive federal safeguards**. Democratic lawmakers have continued to raise concerns that the current framework from the Trump administration prioritizes preemption without establishing sufficiently robust federal safeguards. Proposals such as H.R. 8031, which was introduced on March 20 in the House and which seeks to repeal the Trump administration's preemption efforts on AI, underscore the extent of this opposition. Taken together, these dynamics suggest that, despite increasing legislative activity, there remains no clear consensus on either the scope or structure of federal AI regulation. Despite increased legislative activity, the path forward for federal AI regulation remains unclear due to competing proposals, jurisdictional complexity, and ongoing divisions, particularly over preemption, across and within both parties.

## **A pause buys policymakers time to take stock and design a unified national framework; the status quo locks in conflict**

**Etheridge '25**, Tammi S. Etheridge, 6-10-2025, "State Approaches to AI Regulation Are a Patchwork," International Center for Law & Economics, <https://laweconcenter.org/resources/state-approaches-to-ai-regulation-are-a-patchwork/> //RonitS

**The cumulative effect of these statutes is a patchwork in which legal ambiguity, duplication, and enforcement complexity are the norm. Administrative costs multiply, firms lose flexibility, and focus shifts from innovation to compliance.** These are **textbook examples of regulatory deadweight loss and compliance arbitrage across jurisdictions.**

**A federal moratorium on new state AI regulations would temporarily freeze this dynamic, preserving policymakers' ability to take stock, assess implementation challenges, and design a unified national framework that aligns with economic efficiency, supports research and entrepreneurship, and addresses algorithmic risks in proportion to their potential harms.** A thoughtful federal statute framed around risk assessments, transparency, accountability, and innovation incentives would avoid redundancies and amplify clarity. In contrast, the current wave of state-by-state bills risks locking in conflict, fueling uncertainty, and constraining a domain that thrives on speed and scale.

## **Else, the patchwork tanks the sector: state tech-law patchworks cost over \$1 trillion, with \$200 billion hitting small businesses**

**Castro '22**, Daniel Castro, Luke Dascoli, Gillian Diebold, 1-24-2022, "The Looming Cost of a Patchwork of State Privacy Laws," Information Technology & Innovation Foundation, <https://itif.org/publications/2022/01/24/looming-cost-patchwork-state-privacy-laws/> //RonitS

In the absence of a federal privacy law, a growing **patchwork** of state **laws** burdens companies with multiple, duplicative compliance costs. The **out-of-state costs from 50 such laws could exceed \$1 trillion** over 10 years, **with** at least **\$200 billion hitting small businesses.**

KEY TAKEAWAYS Since 2018, 34 states have passed or introduced 72 privacy bills regulating the commercial collection and use of personal data. These laws create significant compliance costs for in-state businesses and confusion for consumers while also raising costs for out-of-state businesses that increasingly find themselves subject to multiple, duplicative rules. State privacy laws could impose out-of-state costs of between \$98 billion and \$112 billion annually. Over a 10-year period, these out-of-state costs would exceed \$1 trillion. Small businesses would bear \$20–23 billion of this out-of-state burden annually. Congress should pass federal privacy legislation that preempts states, protects consumers, and promotes innovation.

## **Subpoint B: Stronger Grids**



## AI is set to overload the electricity grid: NERC calls AI power swings a “high likelihood, high impact” risk of cascading outages, and construction outpaces grid upgrades

**Behr '26**, Peter Behr, Senior Energy Reporter for Energywire, 3-18-2026, “AI power demand creates ‘high likelihood, high impact’ grid risks,” E&E News, <https://www.eenews.net/articles/ai-power-demand-creates-high-likelihood-high-impact-grid-risks/>, accessed 7-28-2026 // leon + BZ + Willie T. + @circuitskillbuilder

Nationally, **developers** have **announced** \$600 billion in **new hyperscale data center** activity **this year**.

North America’s electric grid watchdog is acting with urgency to guard against widespread power outages stemming from the volatile energy demand of data centers.

The North American Electric Reliability Corp., the grid’s not-for-profit security monitor, is moving to draft new standards for large artificial intelligence computing hubs that could lead to regulation of their operations. A NERC committee is expected to initiate the project Wednesday, seeking final approval by the end of the year.

A recent **NERC white paper** says the **extreme power fluctuations** during the training of AI models, when power demand can swing by **hundreds of megawatts** in an instant, is a ‘**high likelihood, high impact**’ risk. It’s a threat that could potentially throw **carefully calibrated** power grids out of balance and lead to **uncontrolled, cascading outages**.

The report lists “critical reliability gaps” caused by large data centers, calling it “imperative” to address this and other risks.

The step toward regulating computing hubs that are increasing in size and number marks an important new challenge for the AI industry. Google, Amazon, Meta, Microsoft, Anthropic and others are developing gigawatt-scale data center complexes — requiring as much power as a midsize or large American city. “The emergence of gigawatt-scale loads may create exponential complications for study, planning and operations,” the paper found.

**AI’s exponential growth** could **push** total AI peak **power** requirements to 50 gigawatts by **2030**, according to an analysis by the Electric Power Research Institute. Concern that existing power demand and the projections are driving up utility bills for households and other businesses has shaped debate about the AI buildout. **Risks to a U.S. power grid already under stress from extreme weather** are emerging for the broad array of **regulators and standard-setting groups** charged with ensuring the lights stay on.

The Edison Electric Institute, representing investor-owned utilities, urged NERC to require data centers to meet the type of engineering standards and protocols governing generators and other major operators on the grid. “They should be subject to similar regulatory requirements,” EEI said in a statement.

“As very large loads like data centers expand, planners and grid operators need clear visibility into how those loads will connect to and operate on the system,” said Todd Snitchler, CEO of the Electric Power Supply Association. It happened in Virginia

NERC highlighted a 2024 incident when a transmission outage in Virginia created a sharp voltage spike. **Controls for dozens of data centers automatically separated from the grid and switched to backup power to protect sensitive computer chips and their cooling systems.**

Grid operators suddenly faced an **unexpected**, roughly **1,500-megawatt** loss in power demand.

The larger grid was able to keep operating. But NERC said the **same incident** involving **a large cluster** of **supersize data centers** would **create** an **unacceptable risk** of **cascading shutdowns** and **widespread outages**.

The organization issued an alert on these issues last September.

Some of the concerns identified by NERC are well known, including the lack of information from data centers and utilities on how the computing hubs are engineered and operated. That limits grid operators’ understanding of potential risks. The issue is **complicated** by the



speed of data center construction, which often outpaces needed grid upgrades and preparations for the centers' power requirements.

NERC also cited "a lack of collaboration and coordination" between data center operators, utilities and grid operators. For example, said the NERC working group, some data centers aren't sharing adequate information about shifts to backup power, which can impact grid operations.

NERC is moving relatively fast to get new standards written before the end of the year. They would then have to be approved by the Federal Energy Regulatory Commission.

Its accelerated timetable "underscores the urgency that the federal regulator and broader industry are treating the reliability risks posed by large load growth," said Jiecheng "Jeff" Zhao, chief engineer of Elevate Energy Consulting in a recent blog reporting on a NERC technical conference last month. "It will likely lead to mandatory requirements" for data centers, Zhao wrote.

A "critical" risk, NERC says, is the rapid power fluctuations of tens of thousands of computer chips as AI models train to understand and process huge amounts of information. As one training phase ends and the results are compiled, the power demands of a large data center may plunge by 300 megawatts in moments and then ramp back up equally suddenly as training resumes.

NERC noted that data center operators typically do not tell grid operators these sudden shifts are coming or share technical data that grid control room engineers require to safeguard their networks.

Extreme power fluctuations spreading to the grid can activate protective automatic controls that close down power equipment or transmission lines. The result can be a chain reaction of equipment shutdowns, "potentially leading to uncontrolled separation and cascading outages," the NERC report said.

When a data center ramps down, to the grid it looks like a big power customer just disappeared, said Julia Matevosyan, associate director and chief engineer of the Energy Systems Integration Group (ESIG), a nonprofit research association of grid experts. ESIG recently issued a report documenting data center risks to the grid that agrees with the FERC task force analysis.

"If it's just one data center, it's fine. But if you have multiples of them in close proximity, and they all get hit — they all 'disappear' — that massively exacerbates the initial loss," Matevosyan said in an interview.

### Ramp up, ramp down

In addition, disturbances could trigger oscillations that spread to nearby power plants on the grid, damaging and — in extreme cases — destroying drive shafts of electric turbines near the data center, Matevosyan said. "There's no good understanding of what the limit is" when such oscillations hit, she said.

The dangers of sudden grid fluctuations were highlighted in a blog in October by two senior managers at NVIDIA, the dominant AI chip manufacturer. Joint research signed by 50 scientists from NVIDIA, Microsoft, and OpenAI documents the risk, said NVIDIA's Neeraj Srivastava and Harry Petty.

"These volatile swings — representing hundreds of megawatts ramping up and down in seconds — pose a significant threat to the stability of the utility grid, making grid interconnection a primary bottleneck for AI scaling," they wrote.

"We're at a critical inflection point, where the industry can no longer rely on incremental improvements, and a fundamental architectural shift is required ... capable of managing the power demands of modern AI," they added.

**Affirming solves: the core reliability threat is the speed of the buildout, so a pause lets the grid catch up**

**Kelly '25**, Erin Kelly, 6-13-2025, "Rapid Data Center Development Challenges Grid Reliability, NERC Report Says," NRECA, <https://www.electric.coop/rapid-data-center-development-challenges-grid-reliability-nerc-report-says> //RonitS

Huge, electricity-gobbling data **centers** are being **develop**ed **faster than** the generation and transmission **infrastructure** needed **to support them**—a **dilemma that threatens grid reliability**, according to the 2025 State of Reliability assessment by the North American Electric Reliability Corp.

“A **significant** near-term reliability **challenge** ... **is** the **size and speed at which** large data **centers**, typically **developed** to support the computing needs for AI and cryptocurrency mining, are expanding across the country,” the report says.

Data **centers** are often **built** within **two years**—much **faster than** traditional **industrial loads**, the June 12 NERC report says.

Electric cooperatives are increasingly seeing data centers coming into their rural territories, asking for large amounts of power to flow into their businesses 24 hours a day, seven days a week. Many of them are owned by tech giants, including Google, Amazon and Meta.

## **Specifically, it forces data center companies to invest in the grid and meet consistent standards, then lifts once standards are set: New York proves**

**Hochul '26**, Governor Kathy Hochul, 7-14-2026, “First Statewide Moratorium on New Hyperscale Data Centers Launched by Governor Kathy Hochul,” Office of the Governor of New York, <https://www.governor.ny.gov/news/first-statewide-moratorium-new-hyperscale-data-centers-launched-governor-kathy-hochul> //Ronits

Will Establish a Nation-Leading Regulatory Framework That Protects Ratepayers, the Environment and New Yorkers

One-Year Pause Will Ensure New Yorkers Are Not Paying for Transmission and Infrastructure Build Outs

Will Create New York State Data Center Community Investment Framework, Ensuring Localities Access Can Unlock Millions From New Development

Governor Will Pursue Legislation To Repeal Sales Tax Exemptions for Massive Data Centers Across the State

Governor Kathy Hochul today signed an Executive Order to create the nation’s first **moratorium** on new hyperscale data centers, establishing the **strongest standards for data center development** and **creating a blueprint to support localities**.

The Governor is temporarily pausing State environmental **permits** for up to one year in order to **build** a **nation-leading regulatory framework** that **protects ratepayers**, the **environment**, the **energy grid** and **communities across the state**.

“New York has always been at the forefront of innovation and change but we’ve also always guaranteed that New Yorkers benefit. As data center development threatens to hike up utility bills, deplete our natural resources, and create uncertainty for New Yorkers, it’s my responsibility to take action and lead,” Governor Hochul said. “New York will lead the way in creating the strongest standards in the nation for data center development, ensuring that when companies succeed because of New York, New Yorkers succeed too.”

New York State is experiencing **unprecedented growth in demand for data center development driven by artificial intelligence** and other computing operations. The increase in demand has led to proposals across the state for the construction and operation of data centers that could require massive amounts of energy and water to run and cool thousands of computer servers.

Earlier this year, the Governor directed the Department of Public Service (DPS) to **begin the Energize NY proceeding**, which will require data centers to either **pay more for their energy or supply their own**, allowing the state to **keep energy more affordable** for New Yorkers. As part of that proceeding, the Governor is now also **directing DPS to develop a Generic Environmental Impact Statement (GEIS) for data centers to ensure new data centers coming online are being held to consistent standards**. During the development of this GEIS, which will take up to a year, a moratorium will be in place and the **Department of Environmental Conservation (DEC) will not issue any discretionary permits not already deemed complete**. The state will **use the GEIS to assess the potential environmental impacts of the construction and operation of data centers in the state, including their effect on energy demand, water use and quality, and air quality. Once the State**

finalizes these standards, the moratorium will be lifted, allowing new data center projects to proceed as long as they follow state, zoning code and other local approvals.

The Governor also directed Empire State Development (ESD) to issue a Community Investment Framework (CIF), within 60 days, which will provide clear guidance to local entities to help them negotiate community benefits as part of any large-scale data center deal, including local infrastructure improvements, child care investments, and direct financial support for their community. The CIF will also establish frameworks that provide organized labor a seat at the table and prioritize prevailing wage standards and project labor agreements for data center construction, local hiring, apprenticeships and workforce development to maximize economic benefits. This framework will additionally include a formula to help communities assess where to begin investment negotiations. An outline of the CIF is currently available on ESD's website, and the public is encouraged to submit feedback.

"New York will lead the way in creating the strongest standards in the nation for data center development, ensuring that when companies succeed because of New York, New Yorkers succeed too."

Governor Hochul

Additionally, the Governor is directing DPS to consider creating a New York Grid Acceleration Fund to require data centers to invest in the state's aging grid infrastructure and energy needs so all New Yorkers benefit from responsible development. The fund could support the procurement of new clean energy supply and establishment of an insurance pool to which developers may need to contribute to protect against speculative large loads that create uncertainty and increase costs. DPS will also consider approaches to require data centers to fund new clean electric generation dedicated to their operations, including but not limited to customer-sited distributed energy resources and battery storage.

## **Overall, without a strong grid, AI development is impossible: there is no AI without energy, and data center demand more than doubles by 2030**

**IEA '25**, International Energy Agency, 4-10-2025, "Energy and AI," IEA Special Report, <https://www.iea.org/reports/energy-and-ai/executive-summary> //RonitS

There has been a step change in the capabilities of artificial intelligence (AI), driven by falling computation costs, a surge in data availability and technical breakthroughs. AI is the science of making machines capable of learning to perform tasks that traditionally required human intelligence. AI is emerging as a general-purpose technology, much like electricity. Today, it can generate text and videos, accelerate scientific discovery in fields like medicine or materials science, make manufacturing robots smarter and more productive, drive commercial taxis in complex city landscapes, and detect threats to critical infrastructure.

In the past few years, AI has gone from an academic pursuit to an industry with trillions of dollars of market capitalisation and venture capital at stake. The market capitalisation of AI-related firms in the S&P 500 has grown by around USD 12 trillion since 2022. While there are several uncertainties about its uptake and impact, AI's rapid development and huge potential have made it central to corporate strategies, economic policies and geopolitics.

However, there is no AI without energy; at the same time, AI has the potential to transform the energy sector. Affordable, reliable and sustainable electricity supply will be a crucial determinant of AI development, and countries that can deliver the energy needed at speed and scale will be best placed to benefit. Training and deploying AI models takes place in large and power-hungry data centres. A typical AI-focused data centre consumes as much electricity as 100 000 households, but the largest ones under construction today will consume 20 times as much.

Policy makers and markets have lacked the tools to assess implications The energy sector is therefore at the heart of one of the most important technological revolutions today. However, there is still a lack of understanding of the stakes and implications of this deepening connection between energy and AI. Consistent with its strong track record of identifying and exploring emerging issues in the energy sector, this new International Energy Agency (IEA) special report seeks to fill this gap with the most

comprehensive, data-driven analysis on the topic to date. Based on a new global model and comprehensive dataset of data centre electricity demand, its analysis was also enriched by an in-depth process of consultation with policy makers, the tech sector, the energy industry and other experts.

Data centres account for a small share of global electricity consumption today, but their local impacts are far more pronounced Global investment in data centres has nearly doubled since 2022 and amounted to half a trillion dollars in 2024. This investment boom has led to growing concerns about skyrocketing electricity demand.

Data centres accounted for around 1.5% of the world's electricity consumption in 2024, or 415 terawatt-hours (TWh). The United States accounted for the largest share of global data centre electricity consumption in 2024 (45%), followed by China (25%) and Europe (15%). Globally, data centre electricity consumption has grown by around 12% per year since 2017, more than four times faster than the rate of total electricity consumption. AI-focused data centres can draw as much electricity as power-intensive factories such as aluminium smelters, but they are much more

geographically concentrated. Nearly half of data centre capacity in the United States is in five regional clusters. The sector accounts for substantial shares of electricity consumption in local markets.

**Electricity demand for data centres more than doubles by 2030** Data centre electricity consumption is set to more than double to around 945 TWh by 2030. This is slightly more than Japan's total electricity consumption today. AI is the most important driver of this growth, alongside growing demand for other digital services. The United States accounts for by far the largest share of this projected increase, followed by China. In the United States, data centres account for nearly half of electricity demand growth between now and 2030. By the end of the decade, the country is set to consume more electricity for data centres than for the production of aluminium, steel, cement, chemicals and all other energy-intensive goods combined. Uncertainties widen further after 2030, but our Base Case sees global data centre electricity consumption rising to around 1 200 TWh by 2035.

A diverse range of sources will be needed to meet demand Renewables and natural gas take the lead in meeting data centre electricity demand, but a range of sources are poised to contribute. Half of the global growth in data centre demand is met by renewables, supported by storage and the broader electricity grid. Renewables generation is projected to grow by over 450 TWh to meet data centre demand to 2035, building on short lead times, economic competitiveness and the procurement strategies of tech companies. Dispatchable sources, led by natural gas, also have a crucial role to play, with the tech sector helping to bring forward new nuclear and geothermal technologies as well. Natural gas expands by 175 TWh to meet growing data centre demand, notably in the United States. Nuclear contributes about the same amount of additional generation to meet data centre demand, notably in China, Japan and the United States. The first small modular reactors come online around 2030.

**Data centres are one of several drivers of accelerated electricity demand growth in the Age of Electricity** Data centres account for around one-tenth of global electricity demand growth to 2030, less than the share from industrial motors, air conditioning in homes and offices, or electric vehicles. However, the significance of data centres in driving electricity demand differs by country. Emerging and developing economies are already experiencing rapid electricity demand growth. In these countries, data centres account for around 5% of the increase in electricity demand to 2030. Advanced economies, on the other hand, have seen several decades of essentially stagnant electricity demand. In this group of countries, data centres account for more than 20% of demand growth to 2030, presenting a wake-up call on the need to put the electricity sector on a growth footing again.

## Impact: The AI Economy

**Weak foundations forfeit the economy of the future: AI could grow from a \$2 trillion to a \$15.7 trillion market by 2030**

**Sheehan '22**, Kelley Sheehan, J.D. Candidate at the University of San Diego School of Law, 2022, "How the U.S. Can Make Subject Matter Eligibility More Predictable and Free Artificial Intelligence Innovation," San Diego International Law Journal, <https://digital.sandiego.edu/ilj/vol23/iss2/3/>

Not only is A.I. the future of many industries, but it also has the potential to grow from a \$2 trillion international market to a \$15.7 trillion market by 2030.<sup>188</sup> This statistic alone emphasizes the importance of the industry, and as much of this industry should be kept domestically through more straightforward patent issuing to encourage investment in innovation.